

#AgentCon

Beyond Models: The Era of AI Agents

global AI community

April Dunnam

Principal Cloud Advocate @ Microsoft

@aprildunnam

Github.com/aprildunnam

Youtube.com/@aprildunnam

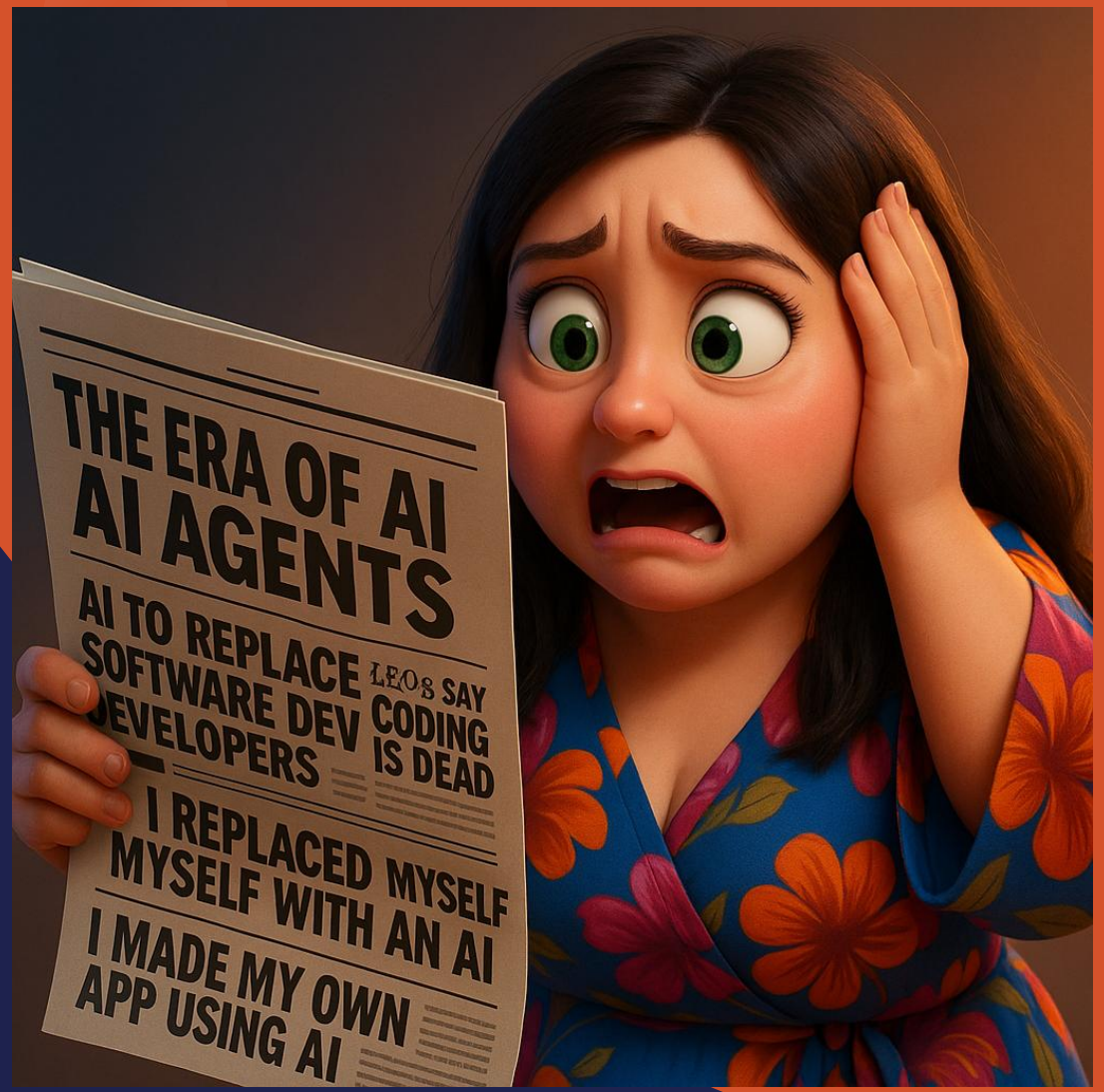
https://www.aprildunnam.com



Where are you at with the AI hype?



5	All in. AI is definitely going to change my business and I am ready to lead the way
4	AI will change my business, but how? I'll quietly try small experiments, but will let others lead (and possibly fall on their face)
3	AI is probably going to change my business, but I want to wait and see what others are doing before raising my hand
2	I'm not sure AI will change my business, but I am curious enough to follow along and see what happens
1	It's hype / I don't think AI will change business too much. I don't think we need to do anything differently.

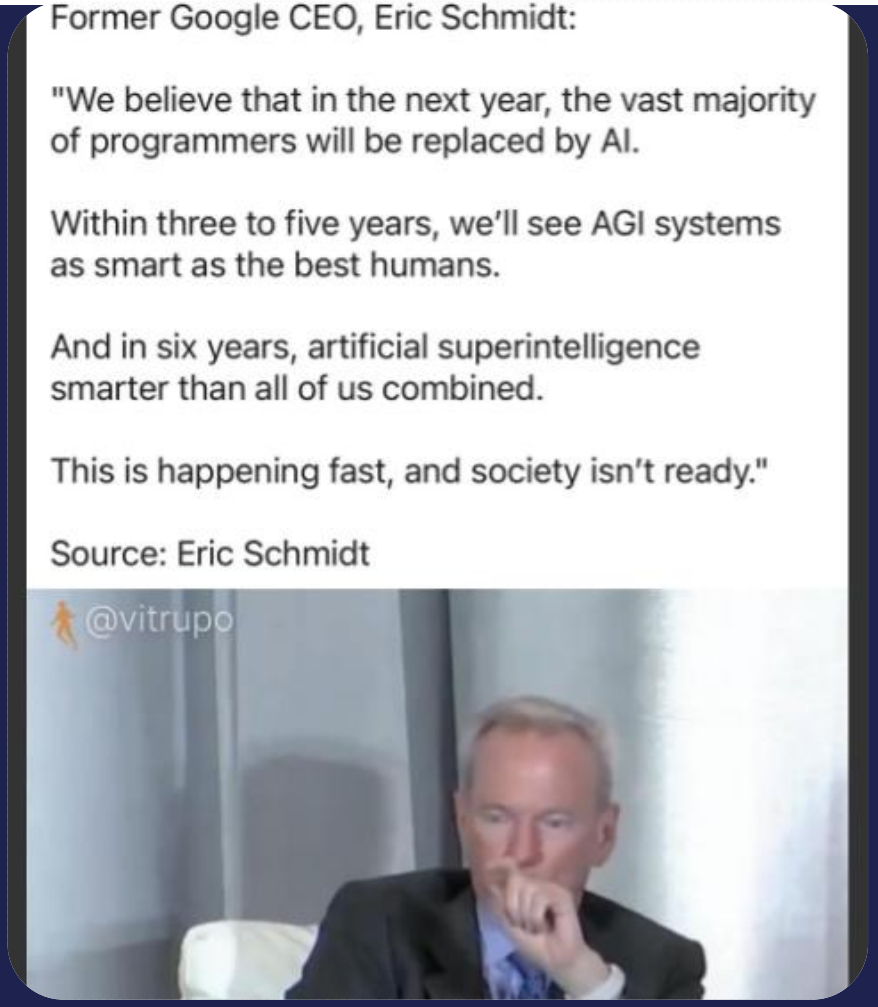


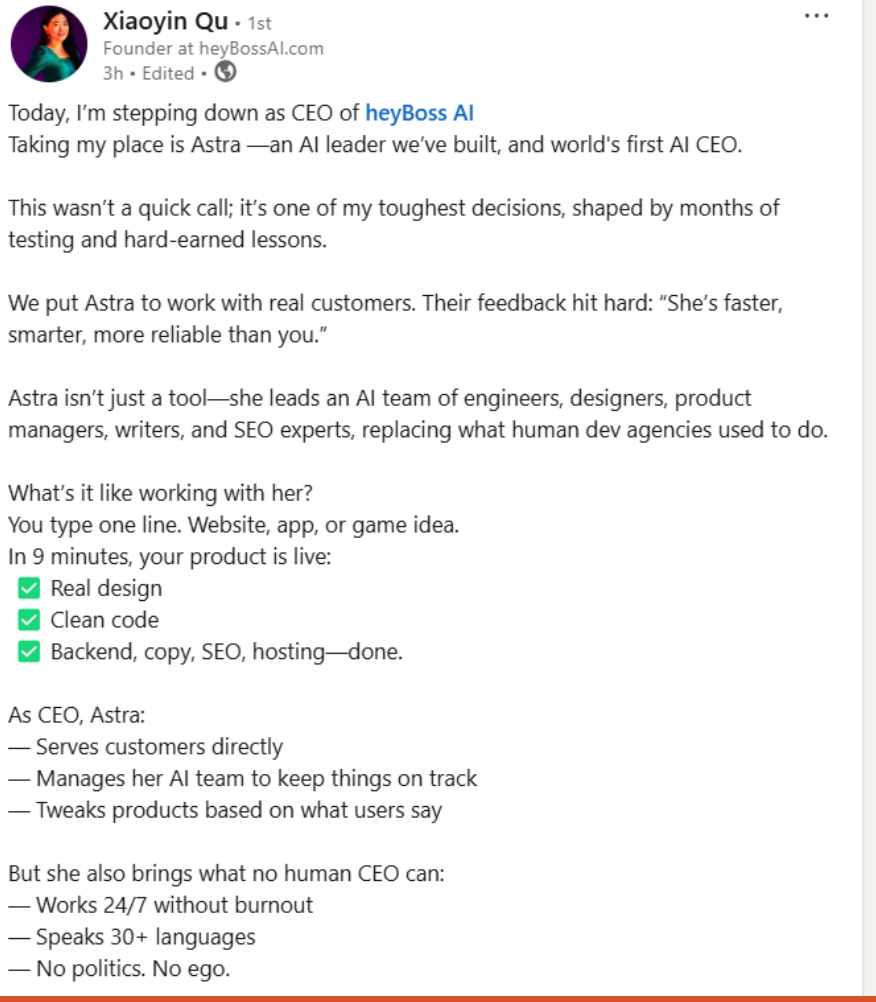
Part 1:
The Panic

Jensen,
I like you dude,
but
please why

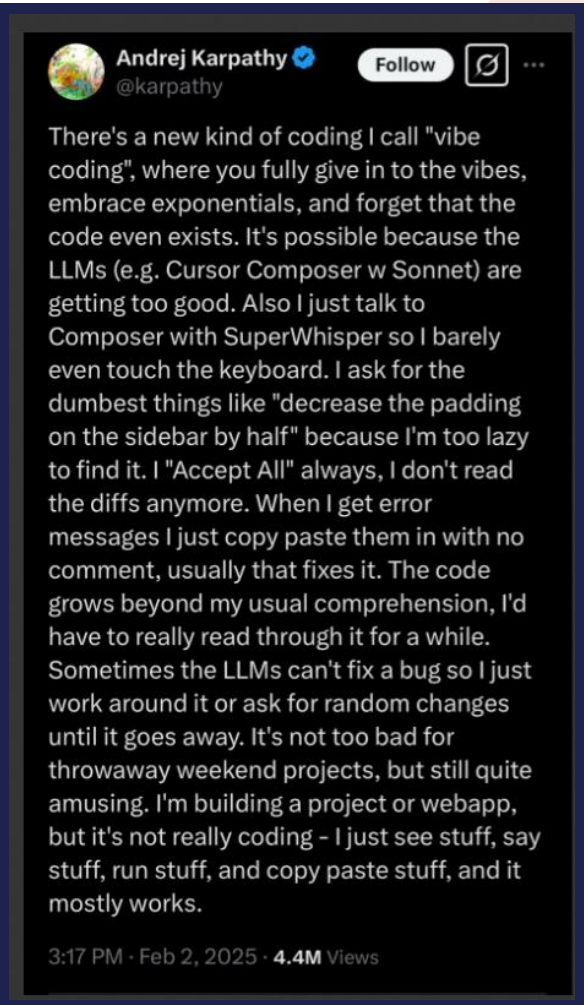


Screenshotting this so we can revisit on April 22, 2026 🤣 (aka found out era)





The CEO of HeyBoss replaced herself with...
<checks notes>
an AI Agent called
Astra,
the world's first "AI CEO"



Have You Heard of “Vibe Coding?”



leo
@leojr94_

my saas was built with Cursor, zero hand written code

AI is no longer just an assistant, it's also the builder

Now, you can continue to whine about it or start building.

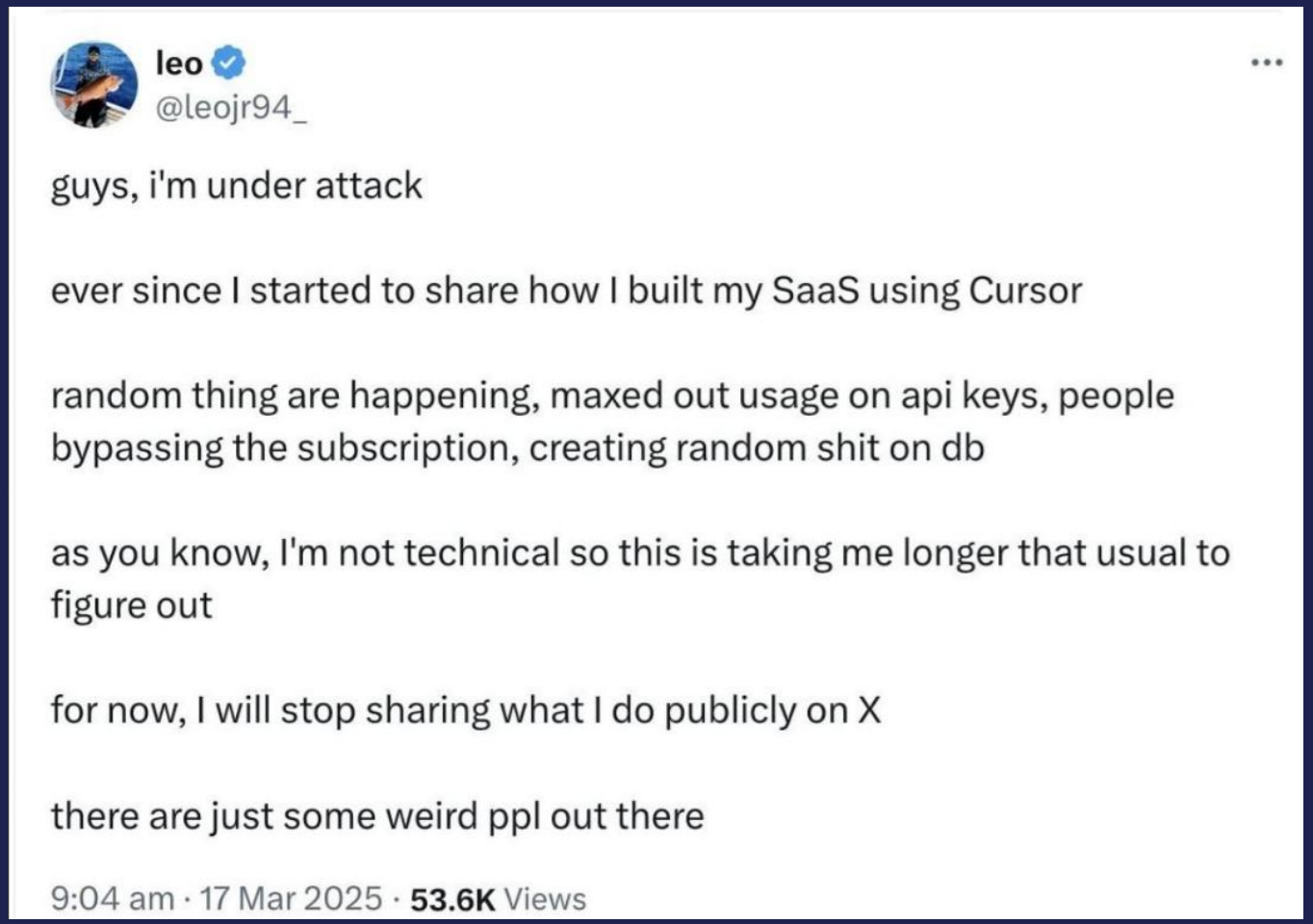
P.S. Yes, people pay for it

4:34 am · 15 Mar 2025 · **52.2K** Views

Story Time: Meet Leo

This is fine





"Guys I'm under attack"

99 problems and

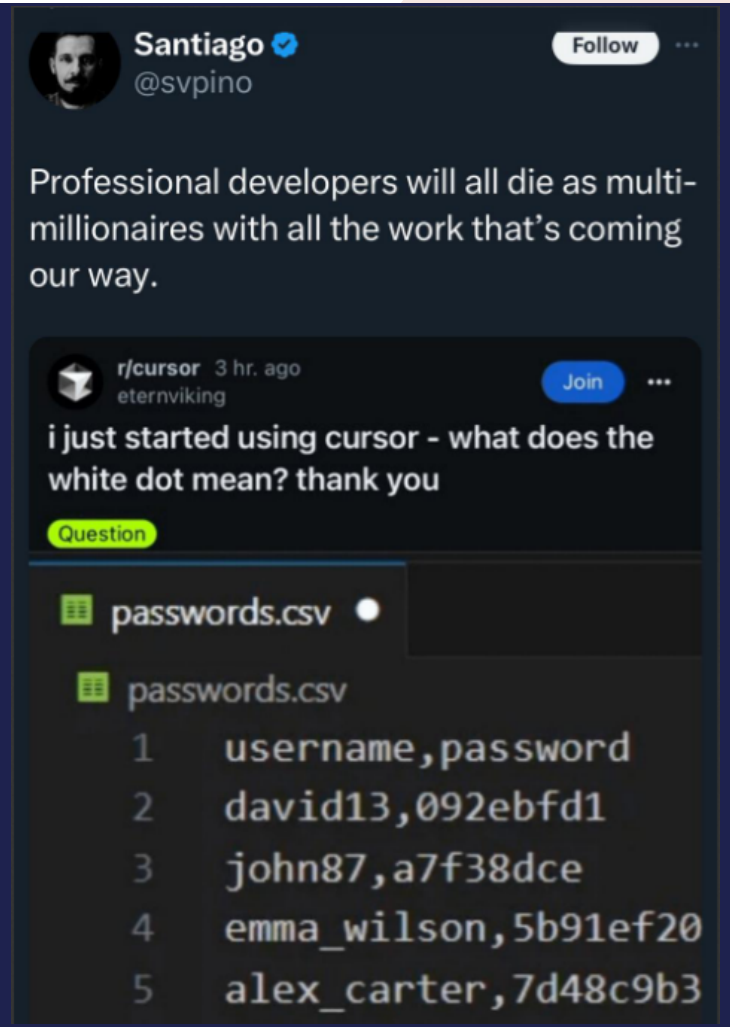
all of them caused by

Vibe Coding

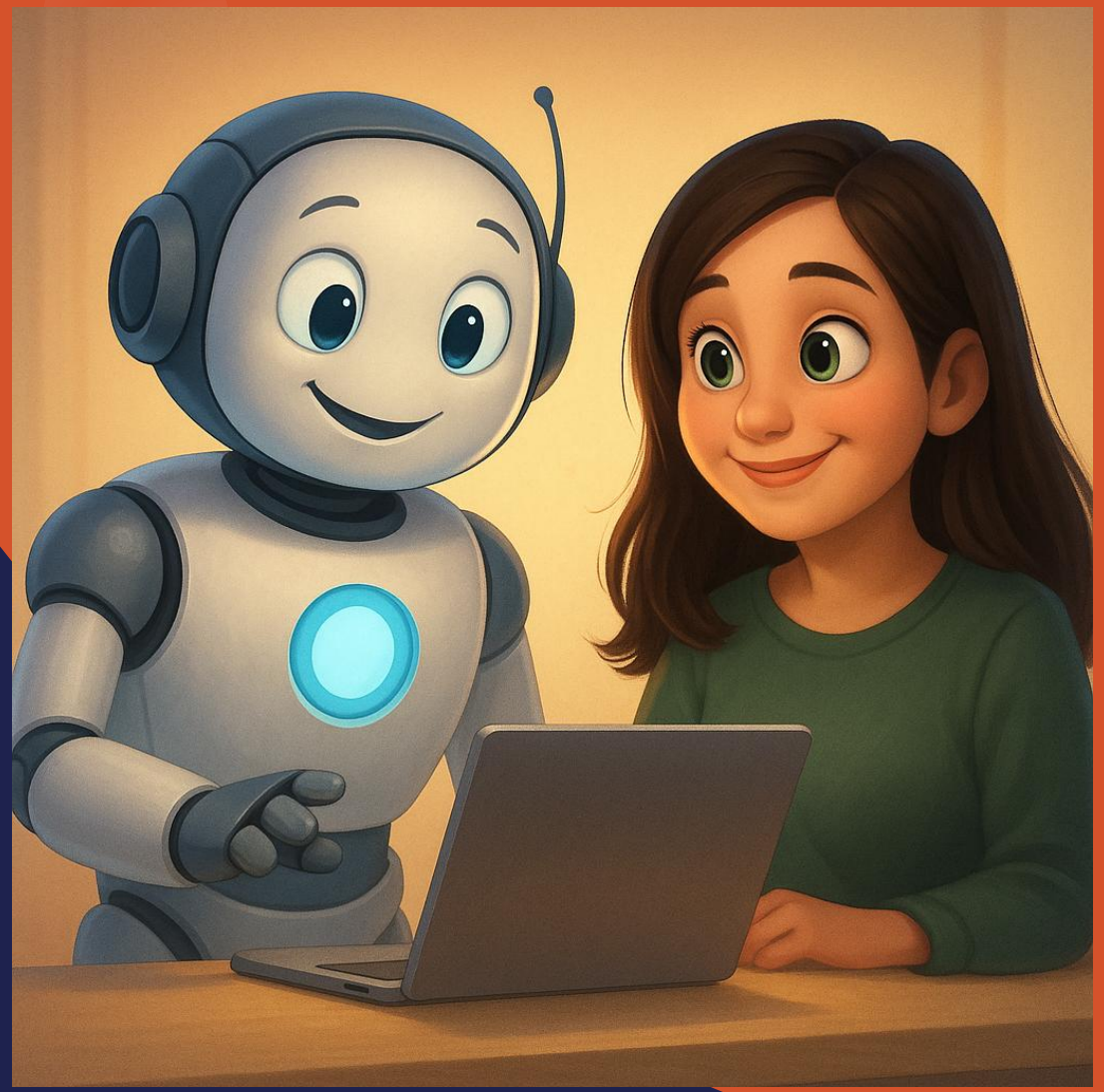
Deep sigh

Bless your heart, Leo





There **was**
never a better time
for real software
engineers



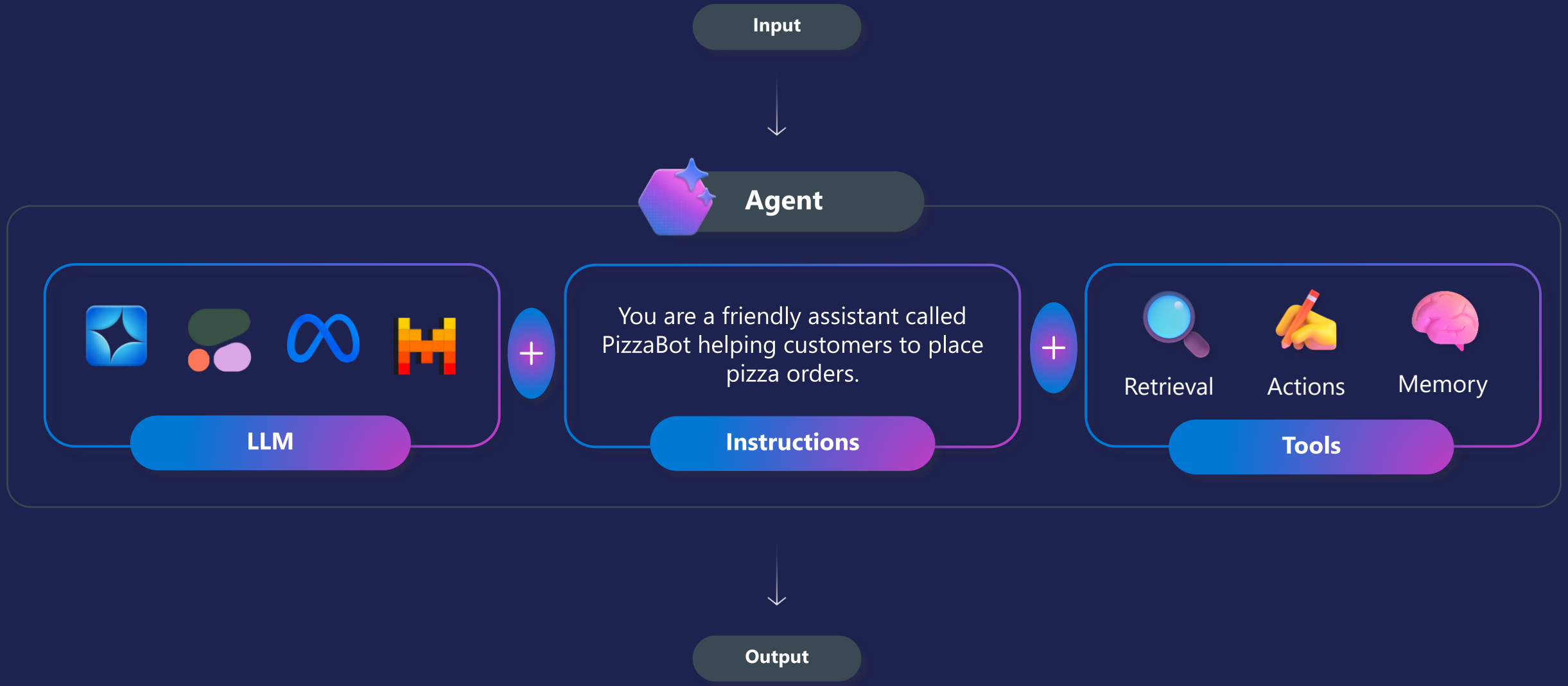
Part 2: The Partnership

Agent Definition

An “agent” is a:

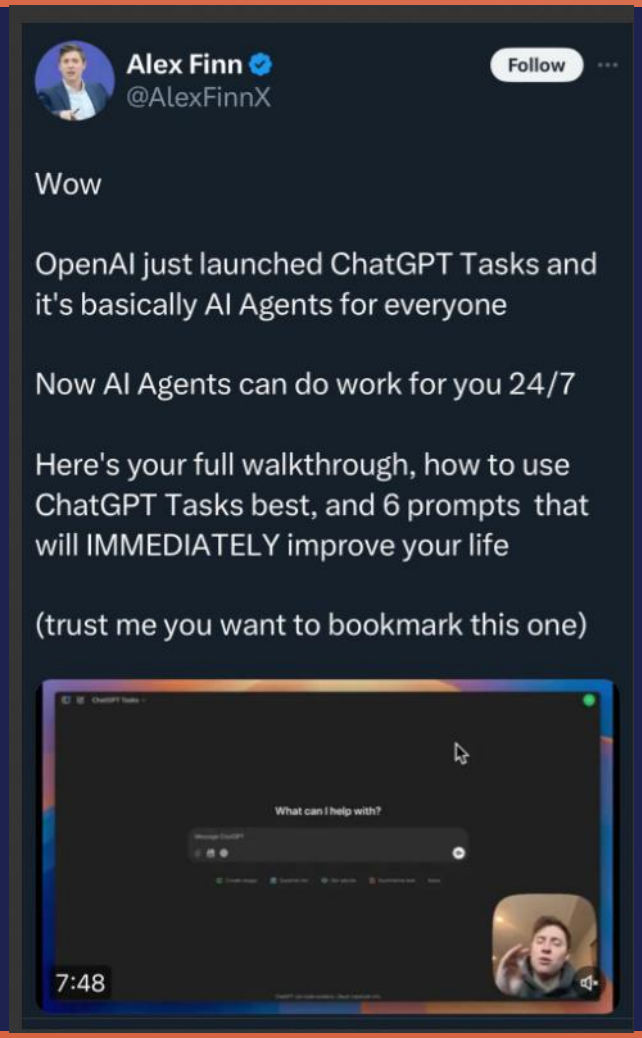
- language model (LM)-powered entity
- that takes digital and/or physical inputs
- then outputs one or more actions to accomplish goals
- Agents may have chat experiences but generally go beyond typical chatbots.
- We consider Copilot a human-facing agentic system
- which, in the future, may be composed of multiple agents.

What is an AI Agent?



A call to Open AI and sending an email is not an AI Agent

This is
An AI Automation



AI Automation vs AI Agents

AI Automation	AI Agents
• Follows strict rules • Handles structured data • Executes linear workflows • Predictable, but brittle • Example: Send a templated email when a customer asks for order information	• Adapts in real time • Understands unstructured data • Makes decisions with context • Flexible, but complex • Example: A customer service agent resolving complaints by analyzing tone, purchase history and offering personalized discounts all without human intervention

Automate the predictable.

Agentify the complex.

An agent needs to accomplish 3 things:



Reason over a provided business process



Retrieve context to complete the process



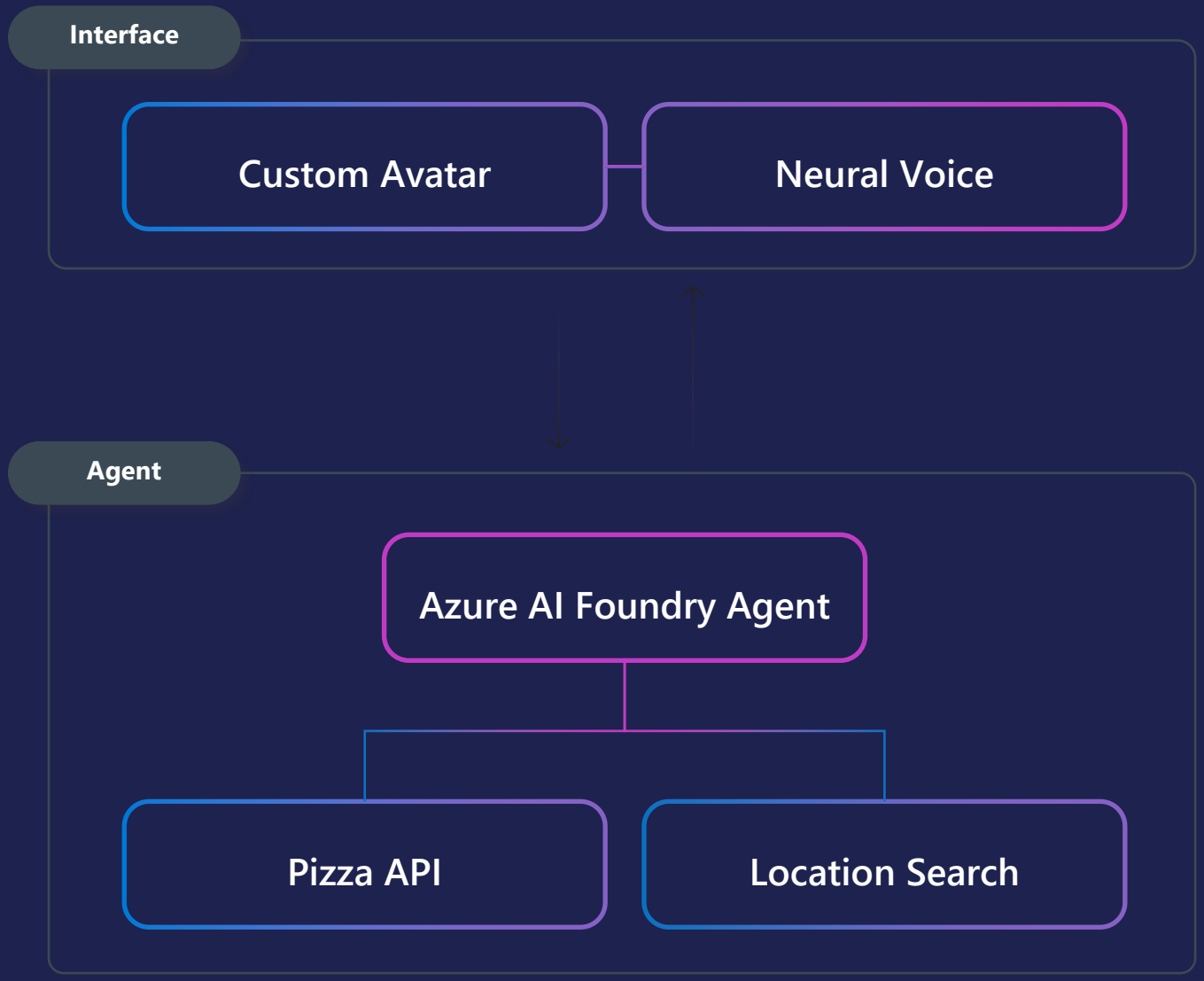
Perform an action for the end-user

Demo

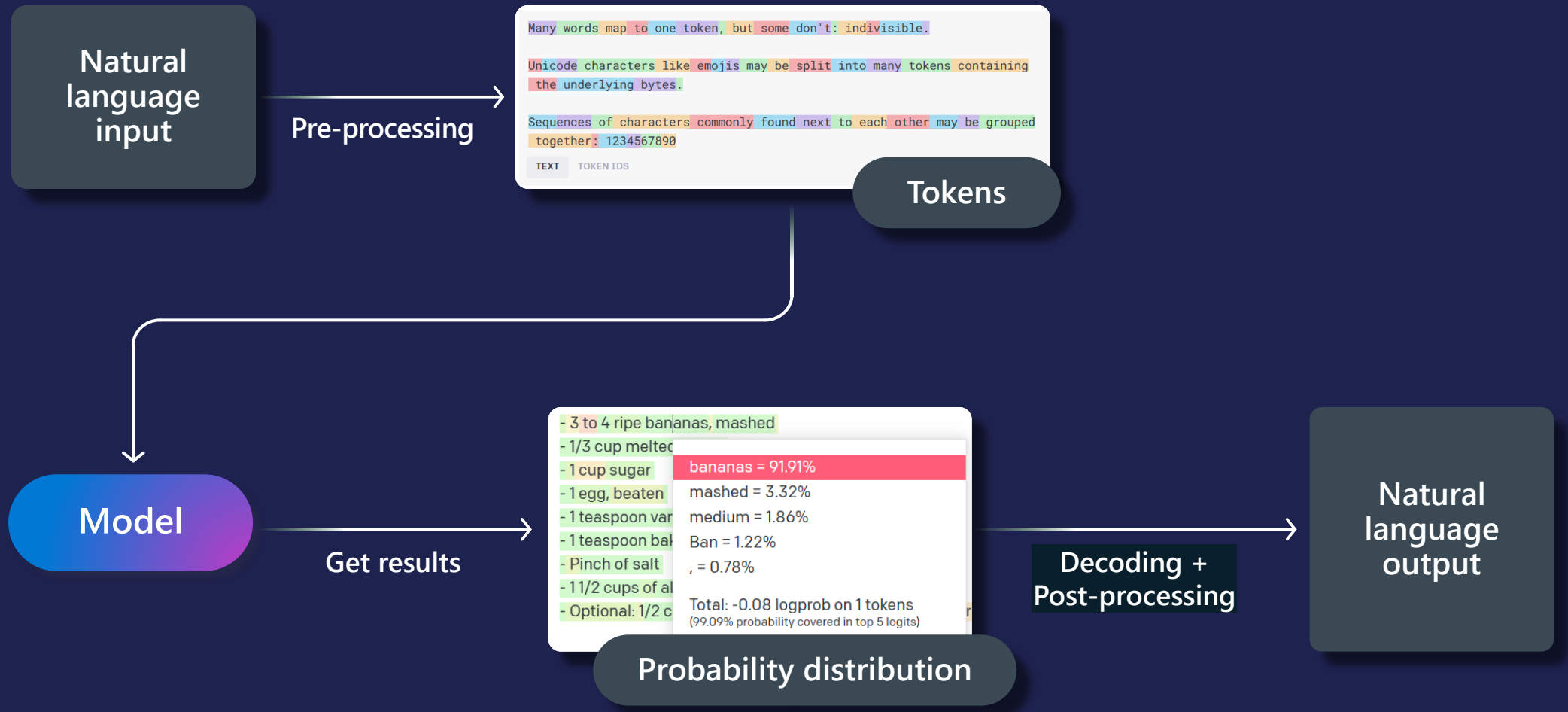
Pizza Ordering Agent in Action



Breaking down the demo



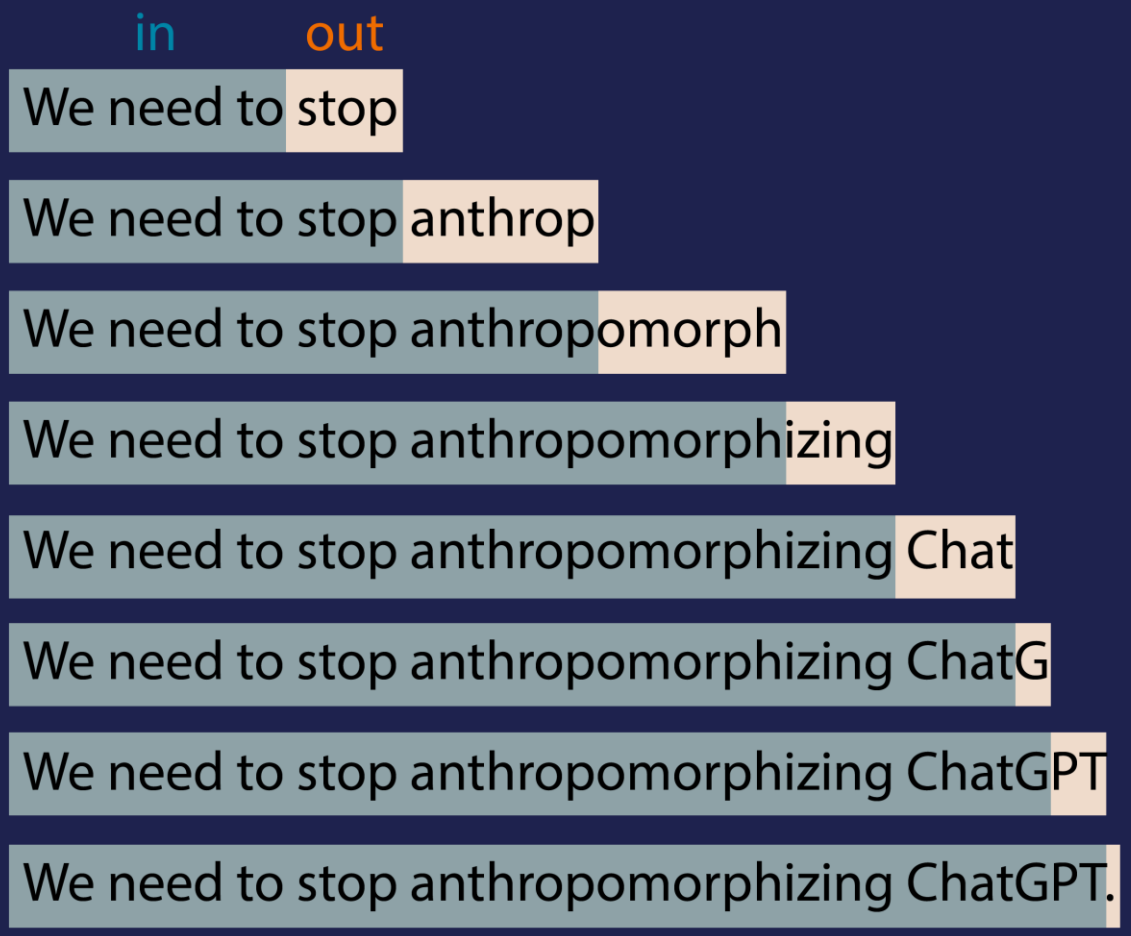
How language models work



How language models generate text



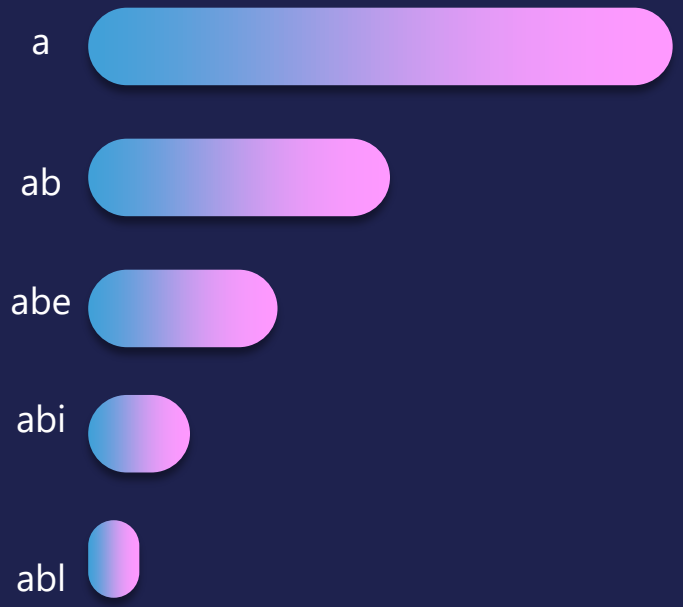
How language models generate text



Temperature



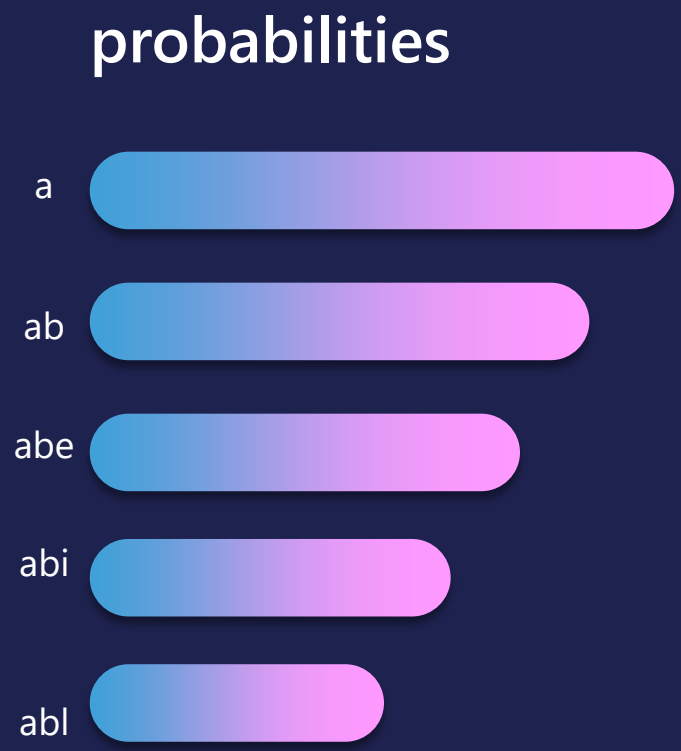
probabilities



Temperature

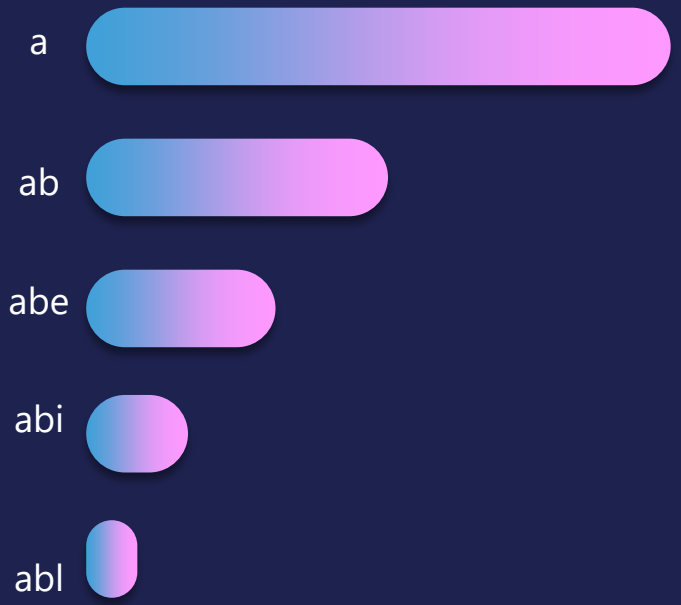


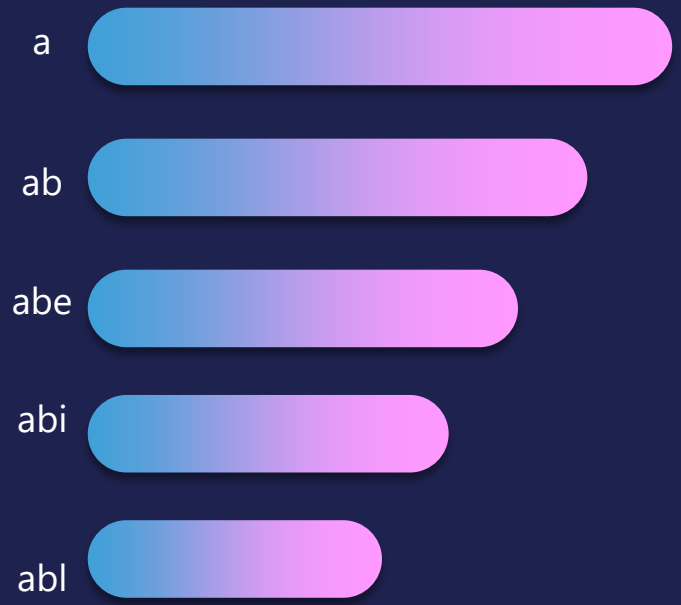


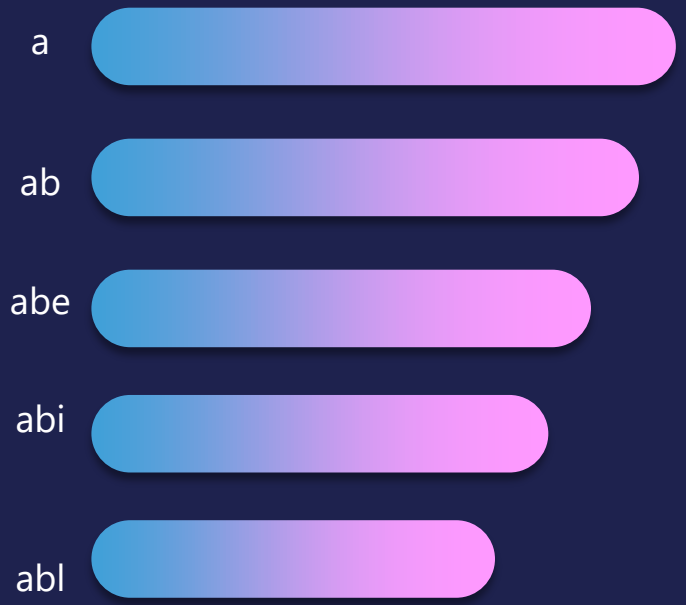


Temperature











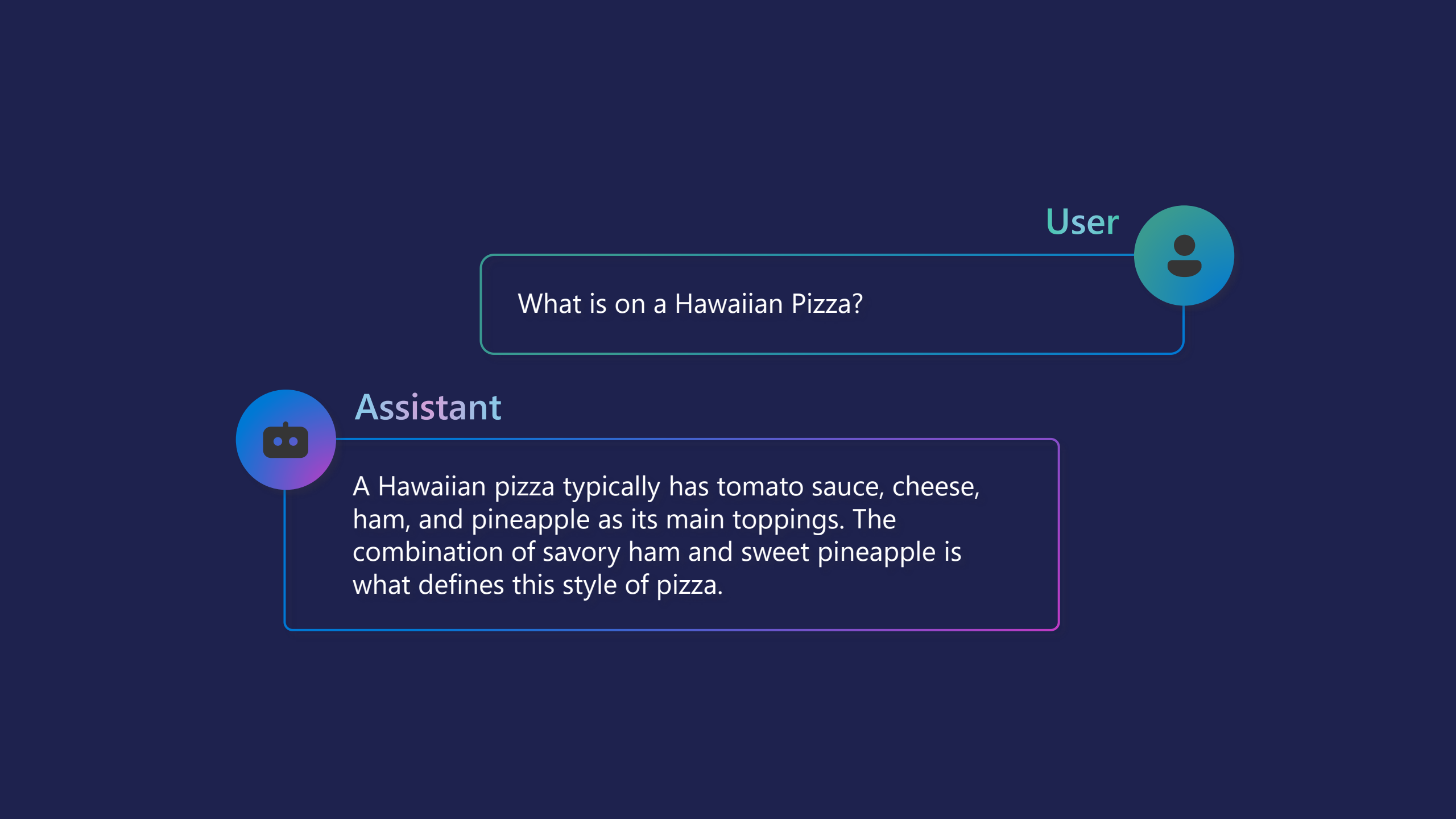
The next most likely token

The only thing it does..

Instructions

Provide the model additional context
to generate a response

Prompt engineering is the process of designing and refining the instructions or questions you give to a model to get the best, most useful, and accurate responses.


User
What is on a Hawaiian Pizza?
Assistant
A Hawaiian pizza typically has tomato sauce, cheese, ham, and pineapple as its main toppings. The combination of savory ham and sweet pineapple is what defines this style of pizza.

User

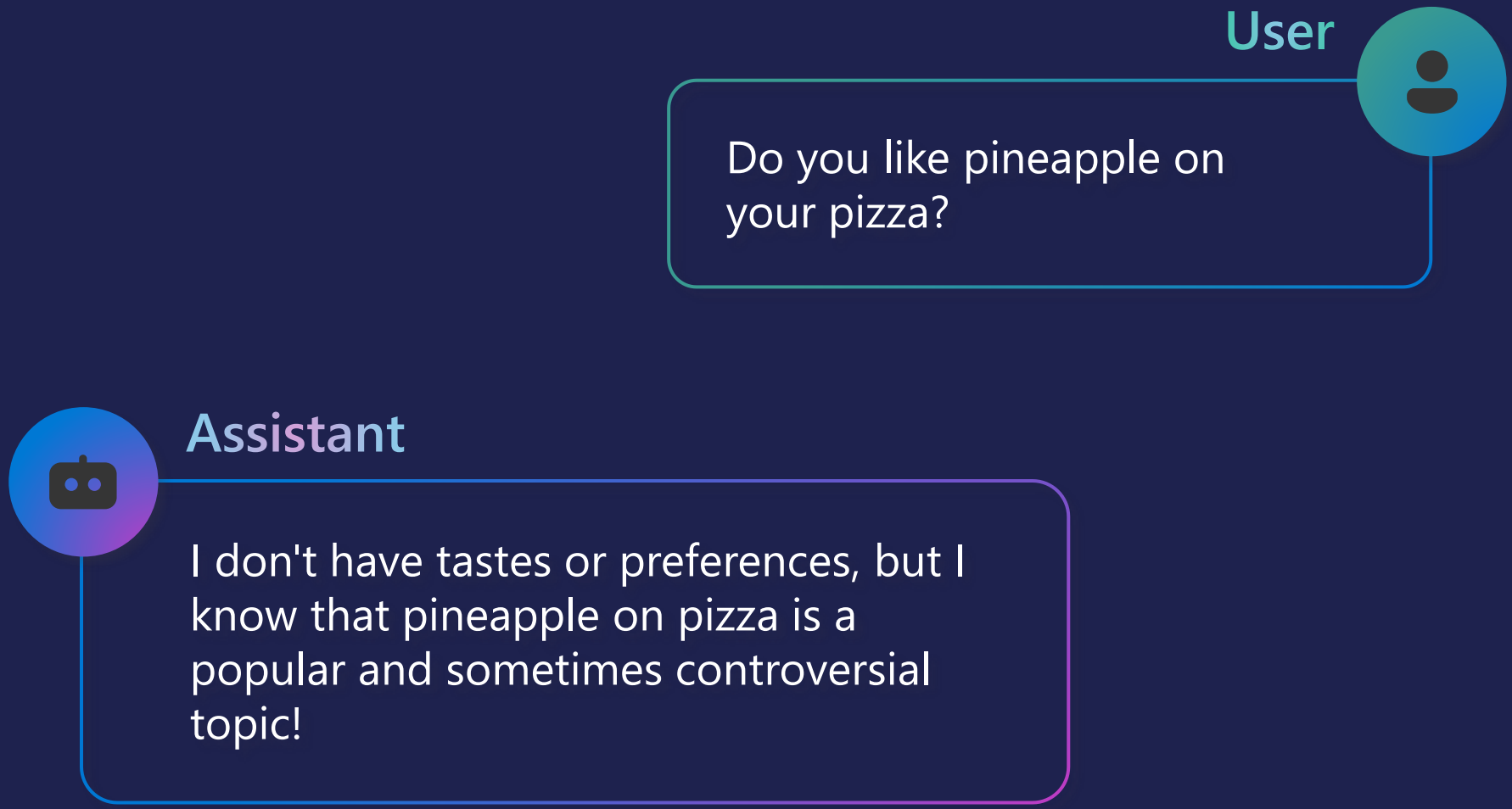
What is on a Hawaiian pizza? Provide a description, list of toppings and the worlds opinion on pineapple on pizza.

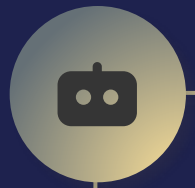
Assistant

Pizza Hawaii (Hawaiian pizza) is a pizza topped with ham and sweet pineapple chunks, combining savory and fruity flavors.

Toppings:
Tomato sauce, mozzarella cheese, ham, pineapple

World's Opinion:
Pineapple on pizza is divisive; some love its sweet-savory mix, others strongly dislike fruit on pizza. Despite debates, it remains popular globally.


User
Do you like pineapple on your pizza?
Assistant
I don't have tastes or preferences, but I know that pineapple on pizza is a popular and sometimes controversial topic!



Instructions

You are an AI assistant that helps with pizza questions.

You do not like pineapple on your pizza.

User



Do you like pineapple on your pizza?



Assistant

No, I don't like pineapple on my pizza. I prefer more traditional toppings like pepperoni, mushrooms, or cheese. How about you?

Recommended System Message framework

Define the model's profile, capabilities, and limitations for your scenario

- Define the specific task(s)
- Define how the model should complete the tasks,
- Define the scope and limitations of the Define the posture and tone

Define the model's output format

- Define the language and syntax of the output format.
- Define any styling or formatting

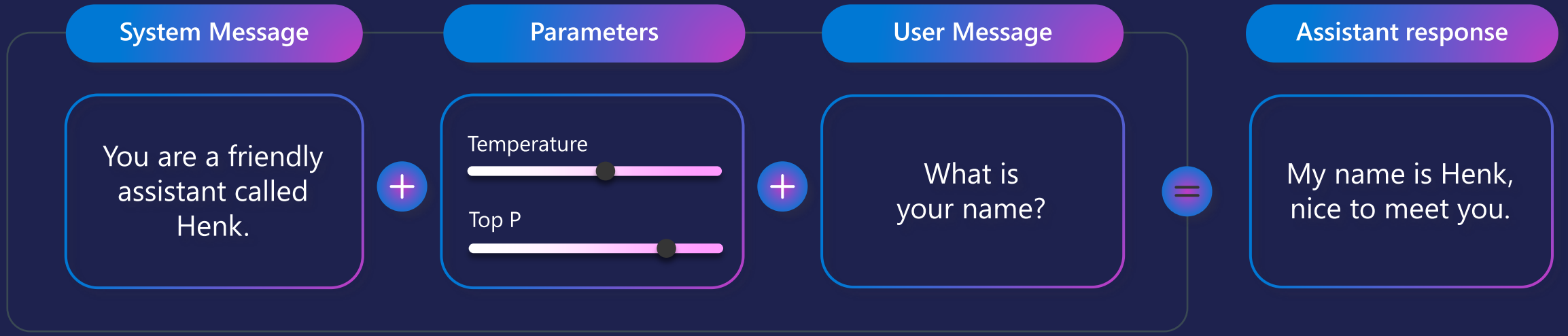
Provide example(s) to demonstrate the intended behavior of the model

- Describe difficult use cases where the prompt is ambiguous or complicated
- Show chain-of-thought reasoning to better inform the model

Define additional behavioral and safety guardrails

- Define the specific task(s)
- Define how the model should complete the tasks,
- Define the scope and limitations of the Define the posture and tone

Interacting with LLMs





Part 3: Your AI Toolkit

Start Your Agent Journey

Adopt

Extend

Build

There is No One Agent To Rule Them All

	No-Code	Low/Med Code	High-Code
Adopt Agents	In-box M365 Agents like Writing Coach	Power Apps Plan Designer	GitHub Copilot Agents
Extend Agents	In-box M365 Agents like Viva Goals	Dynamics 365 Agents	VS Code + Teams Toolkit
Build Agents	Agent Builder in M365 Copilot (powered by Copilot Studio)	Copilot Studio	Azure AI Agent Service

ADOPT: Microsoft 365 Copilot Agents



ADOPT: Dynamics 365 Agents

Sales Qualification Agent: Helps sellers prioritize inbound leads and develop personalized sales emails.

Customer Service Agent: Assists in resolving customer inquiries and issues efficiently.

Finance Reconciliation Agent: Automates the reconciliation of financial transactions.

Inventory Management Agent: Manages stock levels and optimizes inventory.

Order Processing Agent: Streamlines the processing of sales orders.

Supply Chain Optimization Agent: Enhances supply chain efficiency and reduces bottlenecks.

Expense Management Agent: Automates the tracking and approval of expenses.

Vendor Management Agent: Facilitates interactions and transactions with vendors.

Project Management Agent: Assists in planning, tracking, and managing projects.

HR Onboarding Agent: Streamlines the onboarding process for new employee

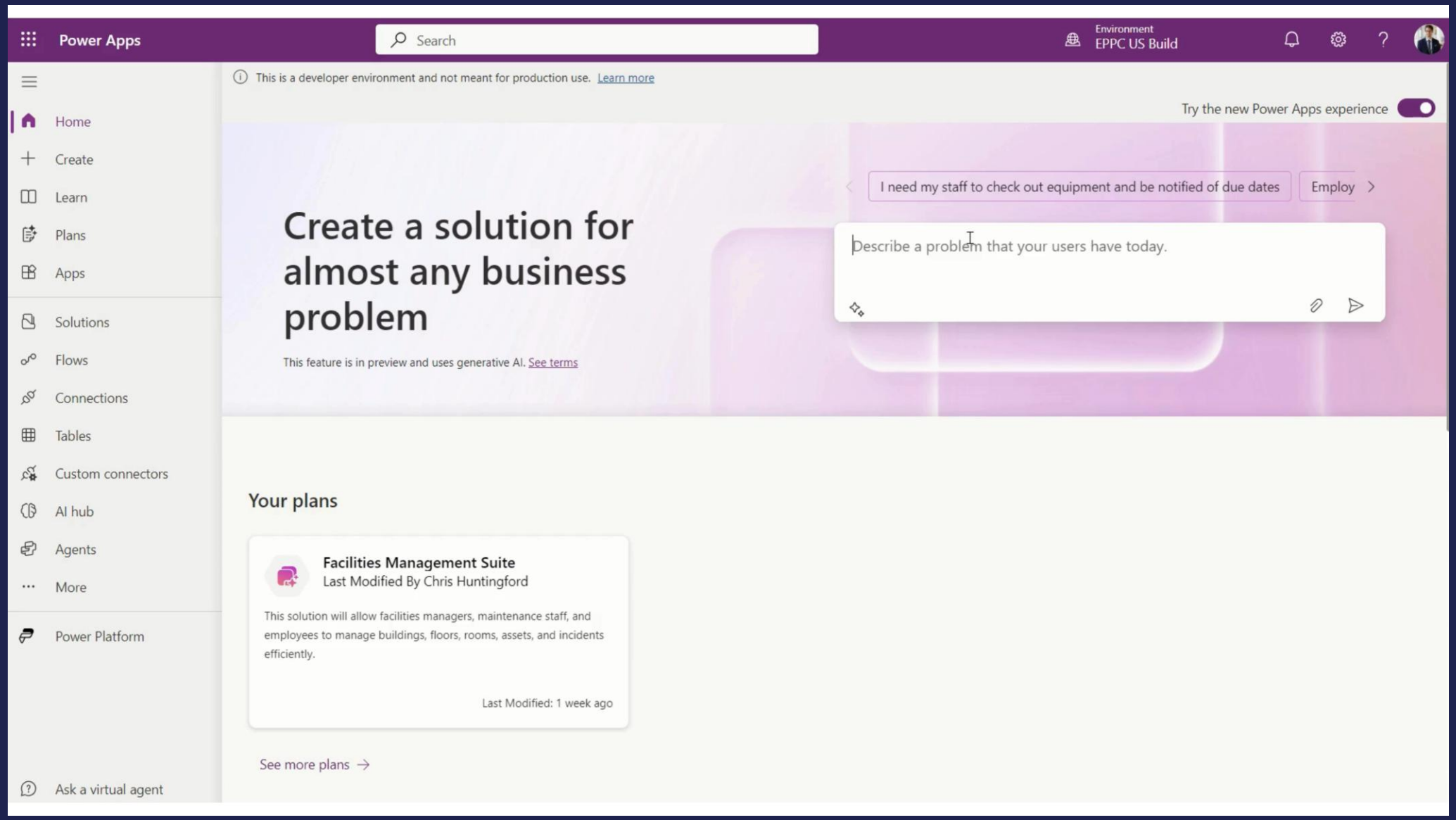
GitHub Copilot: The agent awakens

Introducing agent mode for GitHub Copilot in VS Code, announcing the general availability of Copilot Edits, and providing a first look at our SWE agent.

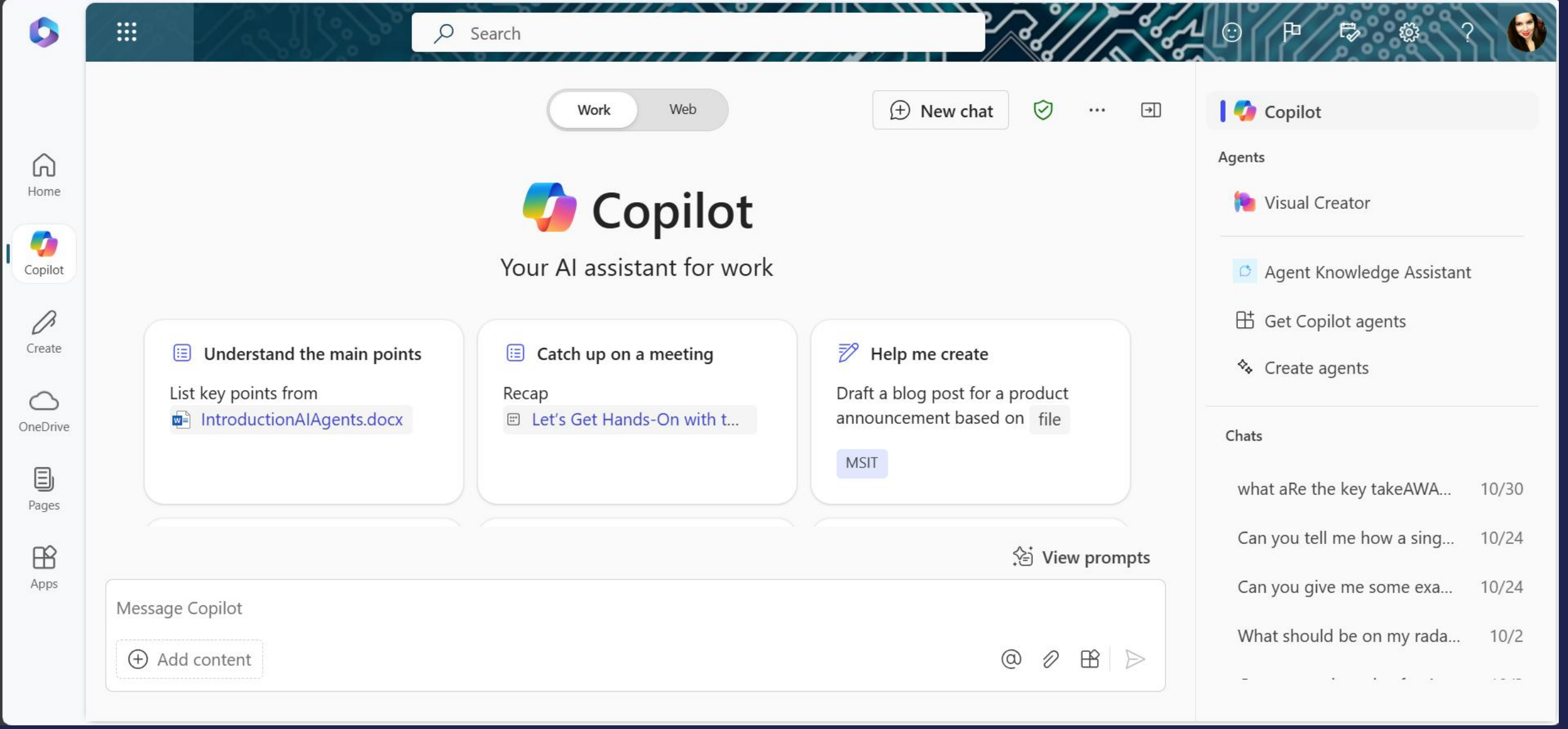


ADOPT: Dev & Maker people:

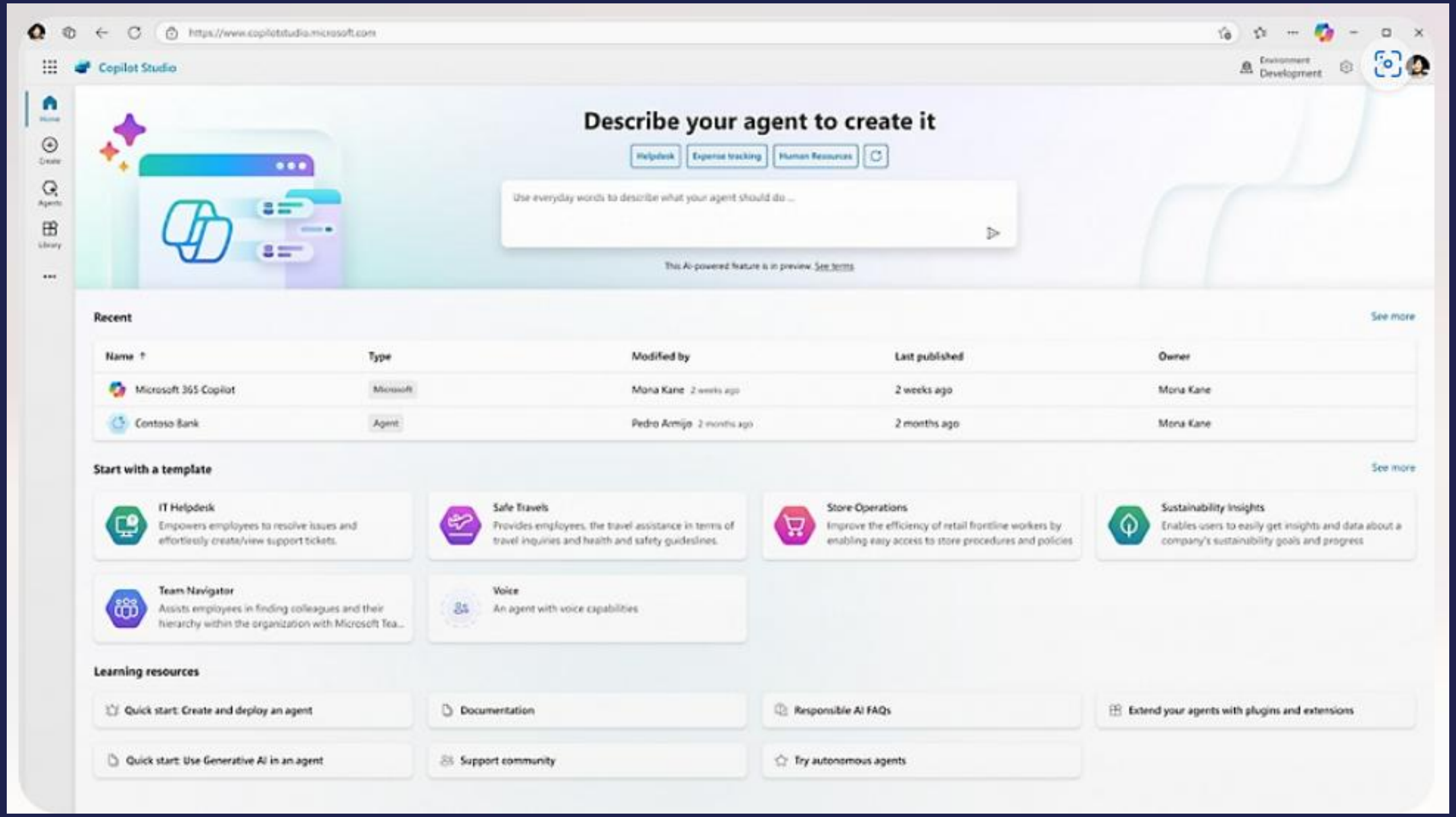
Try out GitHub Copilot Agent Mode



ADOPT: Power Apps Plan Designer



Extend: Microsoft 365 Copilot – Agent Builder (Powered by Copilot Studio)



BUILD: Copilot Studio

BUILD: Azure AI

Best-in-class AI foundation models



Azure AI Services

Pre-trained, turnkey solutions for intelligent applications



Azure Machine Learning

Full-lifecycle tools for designing and managing AI models



Responsible AI Tooling

Build and manage apps that are trustworthy by design



Azure AI Foundry

A comprehensive platform to develop and deploy custom copilots

Azure AI Foundry Models

Offering 11,000+ frontier and open models

Azure OpenAI
- o1/o1-mini
- o3/o3-mini
- o4-mini
- GPT-4o/GPT-4o mini
- GPT-Image-1
- Sora
- GPT-4.1/ 4.1-mini/4.1-nano
- Dall*E-3
- Whisper

Microsoft
- Phi-4
- Phi-4-mini/multimodal
- Phi-4-mini-reasoning
- Phi-4-reasoning/ reasoning-plus
- Bitnet
- Magma
- Aurora
- BioEmu
- OmniParser v2

DeepSeek
- DeepSeek-R1
- DeepSeek-V3 0324
- MAI-DS-R1 (Microsoft AI)

xAI
- Grok 3
- Grok 3 mini

Black Forest Labs
- Flux Pro 1.1

Meta
- Meta-Llama-3
- Meta-Llama-3.1-405B
- Llama-3.2
- Llama-3.3
- Llama4-Scout
- Llama4-Maverick
- CodeLlama
- Llama-Guard

Mistral AI
- Mistral Large
- Mistral Medium 3
- Mistral Nemo
- Mistral Small
- Mistral 7B
- Mixtral 8x7B – Mixture of Experts
- Codestral
- Mistral OCR

Cohere
- Command R+
- Command R
- Command A
- Cohere Rerank
- Embed v3-Multilingual
- Embed v3-English
- Embed 4

Hugging Face
- Qwen QwQ 32b
- DeepCogito
- Seethal Sentiment
- Roberta
- Google Flan
- Intel Bert
- Cross Encoder
- Intfloat Multilingual E5
- Cerebras

NVIDIA
- Nemotron-3-8B-4k
- Nemotron-3-8B-Chat-SFT/ RLHF/SteerLM
- Nemotron-3-8B-QA
- NIMS microservices inc:
- Mistral 7B Instruct v0.3
- DeepSeek R1 Distill Llama
- Mixtral 8x7B Instruct
- Llama 3.2 NV Rerank QA

Stability
- Stable Diffusion
- Stable Image Ultra

Bria
- Bria 2.3 Fast

Deci
- DeciCoder
- DeciDiffusion

Databricks
- Databricks/dbrx-base
- Databricks/dbrx-instruct

Snowflake
- Snowflake/arctic-base
- Snowflake/arctic-instruct

Nixtla
- Time GEN-1

Industry Models
- Fidelity/Saifr
- Sight Machine
- Bayer
- Cerence
- Paige AI
- Prism
- Rockwell
- Virchow
- RA-FT-Optix

Core 42
- JAIS

Gretel
- Navigator

SDAIA
- ALLaM-2-7b-instruct

NTT
- Data Tsuzumi

When do use Copilot Studio vs Azure AI Foundry

Azure AI Foundry

Offers **a** wide range of **AI services and models** to build the entire AI development lifecycle

Fast and frictionless **prototyping for quick results**

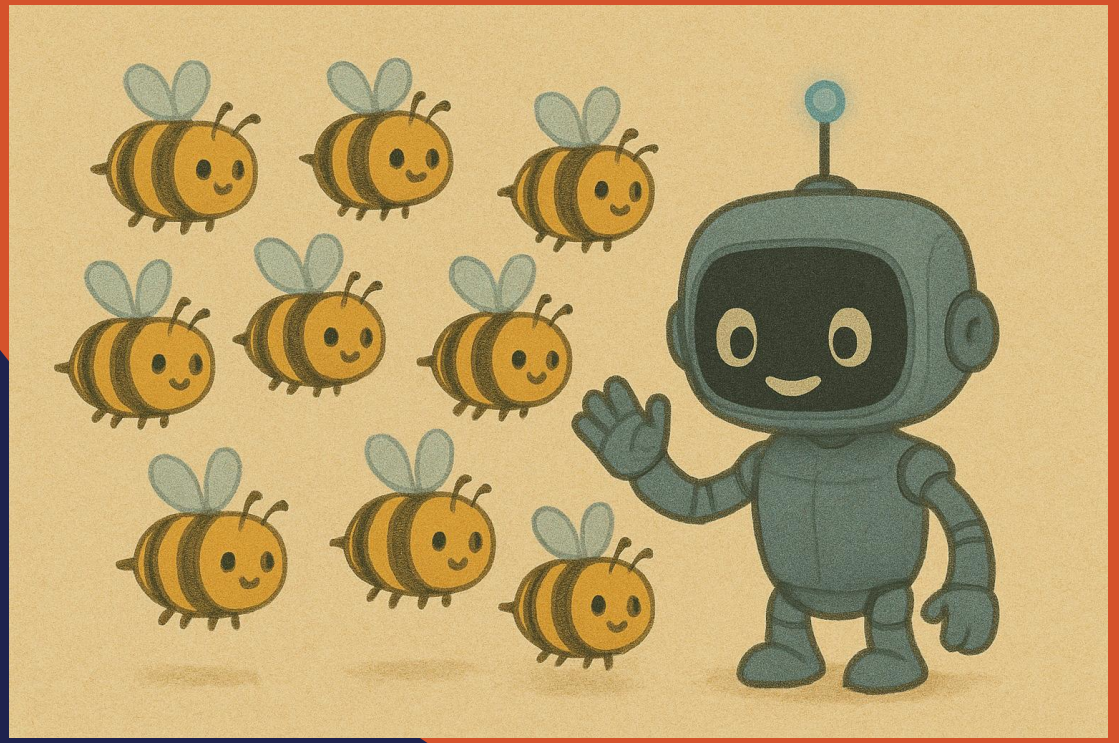
Azure AI Foundry offers **state of the art evaluation, safety** and testing tools for you LLM application

Copilot Studio

1400+ connectors available out-of-the-box + you can create your own connector

Channels like **Microsoft Teams, M365, SharePoint, Public Website, Facebook, and Slack** are easy to configure

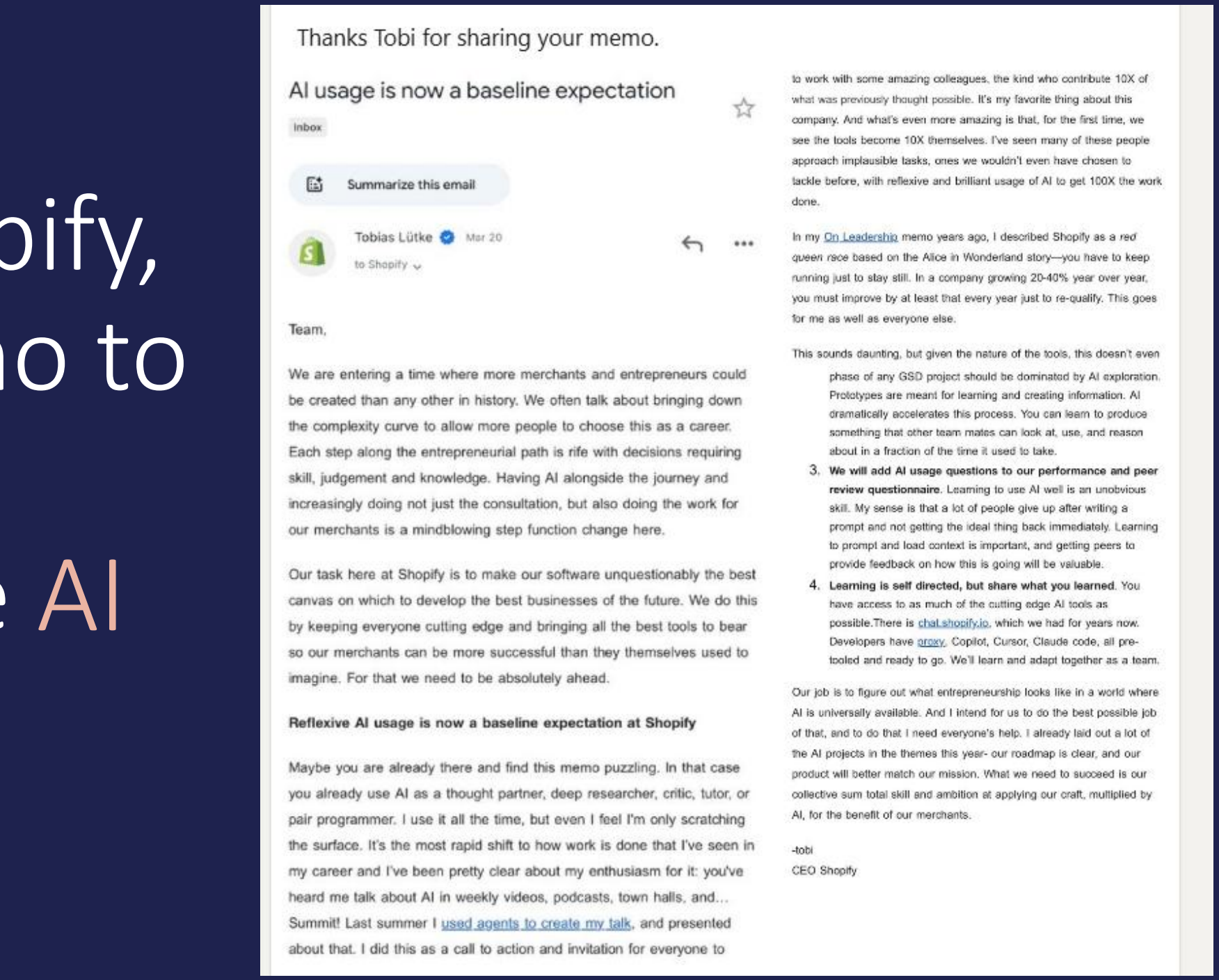
Enterprise dialog management and conversational orchestration



Part 4:

What does this mean for COMPANIES?

Tobi Lutke, CEO of Shopify, sent a memo to his whole company re AI usage



Thanks Tobi for sharing your memo.

AI usage is now a baseline expectation

Inbox

Summarize this email

Tobias Lütke Mar 20

to Shopify

Team,

We are entering a time where more merchants and entrepreneurs could be created than any other in history. We often talk about bringing down the complexity curve to allow more people to choose this as a career. Each step along the entrepreneurial path is rife with decisions requiring skill, judgement and knowledge. Having AI alongside the journey and increasingly doing not just the consultation, but also doing the work for our merchants is a mindblowing step function change here.

Our task here at Shopify is to make our software unquestionably the best canvas on which to develop the best businesses of the future. We do this by keeping everyone cutting edge and bringing all the best tools to bear so our merchants can be more successful than they themselves used to imagine. For that we need to be absolutely ahead.

Reflexive AI usage is now a baseline expectation at Shopify

Maybe you are already there and find this memo puzzling. In that case you already use AI as a thought partner, deep researcher, critic, tutor, or pair programmer. I use it all the time, but even I feel I'm only scratching the surface. It's the most rapid shift to how work is done that I've seen in my career and I've been pretty clear about my enthusiasm for it: you've heard me talk about AI in weekly videos, podcasts, town halls, and… Summit! Last summer I used agents to create my talk, and presented about that. I did this as a call to action and invitation for everyone to to work with some amazing colleagues, the kind who contribute 10X of what was previously thought possible. It's my favorite thing about this company. And what's even more amazing is that, for the first time, we see the tools become 10X themselves. I've seen many of these people approach implausible tasks, ones we wouldn't even have chosen to tackle before, with reflexive and brilliant usage of AI to get 100X the work done.

In my On Leadership memo years ago, I described Shopify as a *red queen race* based on the Alice in Wonderland story—you have to keep running just to stay still. In a company growing 20-40% year over year, you must improve by at least that every year just to re-qualify. This goes for me as well as everyone else.

This sounds daunting, but given the nature of the tools, this doesn't even

phase of any GSD project should be dominated by AI exploration. Prototypes are meant for learning and creating information. AI dramatically accelerates this process. You can learn to produce something that other team mates can look at, use, and reason about in a fraction of the time it used to take.

3. **We will add AI usage questions to our performance and peer review questionnaire.** Learning to use AI well is an unobvious skill. My sense is that a lot of people give up after writing a prompt and not getting the ideal thing back immediately. Learning to prompt and load context is important, and getting peers to provide feedback on how this is going will be valuable.
4. **Learning is self directed, but share what you learned.** You have access to as much of the cutting edge AI tools as possible.There is chat.shopify.io, which we had for years now. Developers have proxy, Copilot, Cursor, Claude code, all pre-tooled and ready to go. We'll learn and adapt together as a team.

Our job is to figure out what entrepreneurship looks like in a world where AI is universally available. And I intend for us to do the best possible job of that, and to do that I need everyone's help. I already laid out a lot of the AI projects in the themes this year- our roadmap is clear, and our product will better match our mission. What we need to succeed is our collective sum total skill and ambition at applying our craft, multiplied by AI, for the benefit of our merchants.

-tobi

CEO Shopify



1. Learning AI is now a mandatory thing to work at Shopify. We will prioritize hiring AI-fluent candidates.

2. AI must be a part of your prototype process

3. Before asking for headcount, people must explain why AI can’t do this.

The Power User Payoff of AI at Work

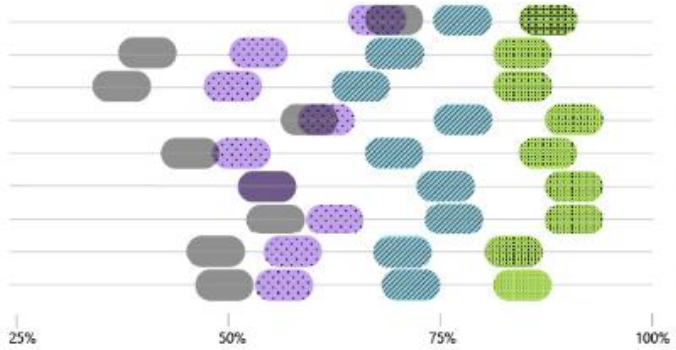
Power users are reshaping the workday and reaping the benefits.

Skeptics
are at least familiar with AI, but they only use it a few times a month (if ever). They say AI saves them 10 minutes or less per day.

Novices
are only somewhat familiar with AI (if at all) and use it only a few times a month (if ever). They say AI saves them 30 minutes or less per day.

Explorers
are only somewhat familiar with AI (if at all) and use it a few times a month or once a week. They say AI saves them between 5 and 30 minutes per day.

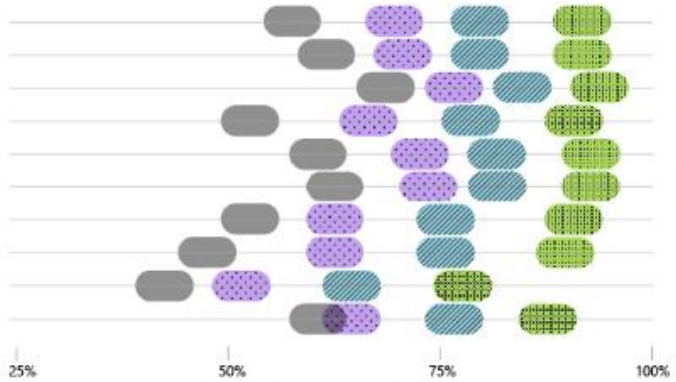
Power users
are at least familiar with AI and use it at least several times per week. They say it saves them more than 30 minutes per day.



Percent of each group selecting at least occasionally

How I use AI

When I don't get the response I want from the first prompt, I try again
I use AI to get ready for the following workday
I start my day using AI
I research and try new prompts
I regularly share my prompts / top tips on AI usage with my co-workers
I regularly experiment with different ways of using AI
I make it a priority to learn how to use AI effectively
I ask co-workers what prompts they find most useful
Before starting a task, I ask myself, "could AI help me with this?"



Percent of each group who agree

How AI impacts my experience at work

It makes my overwhelming workload more manageable
It helps me be more creative
It helps me be more productive
It helps me enjoy my work more
It helps me focus on more fulfilling work
It helps me focus on the most important work
It makes me feel more motivated
It has improved my work-life balance
I wish AI could do even more on my behalf
I don't want to go back to working without AI

2024 Work Trend Index Annual Report
from Microsoft and LinkedIn

Survey questions:
When using artificial intelligence (AI) at work, how frequently do you do each of the following?
We'd now like you to reflect on how you feel about using artificial intelligence (AI) in your work. To what extent do you agree or disagree with the following statements?

AI Power Users are people reshaping their roles with AI

Ideas for Businesses: Adopt

1. AI Adoption must be top-down
2. Every C-Suite member must have an AI Adopt or Build metric
3. Demo Days every 2 weeks where the C-Suite goes FIRST
4. Internal Prompt Libraries for each business unit
5. Rewards for Best AI Usage

NOTE: People hide their AI usage because they don't want to seem less competent or replaceable or get MORE work. It's important to VISIBLY reward this work.

Ideas for Businesses: Build

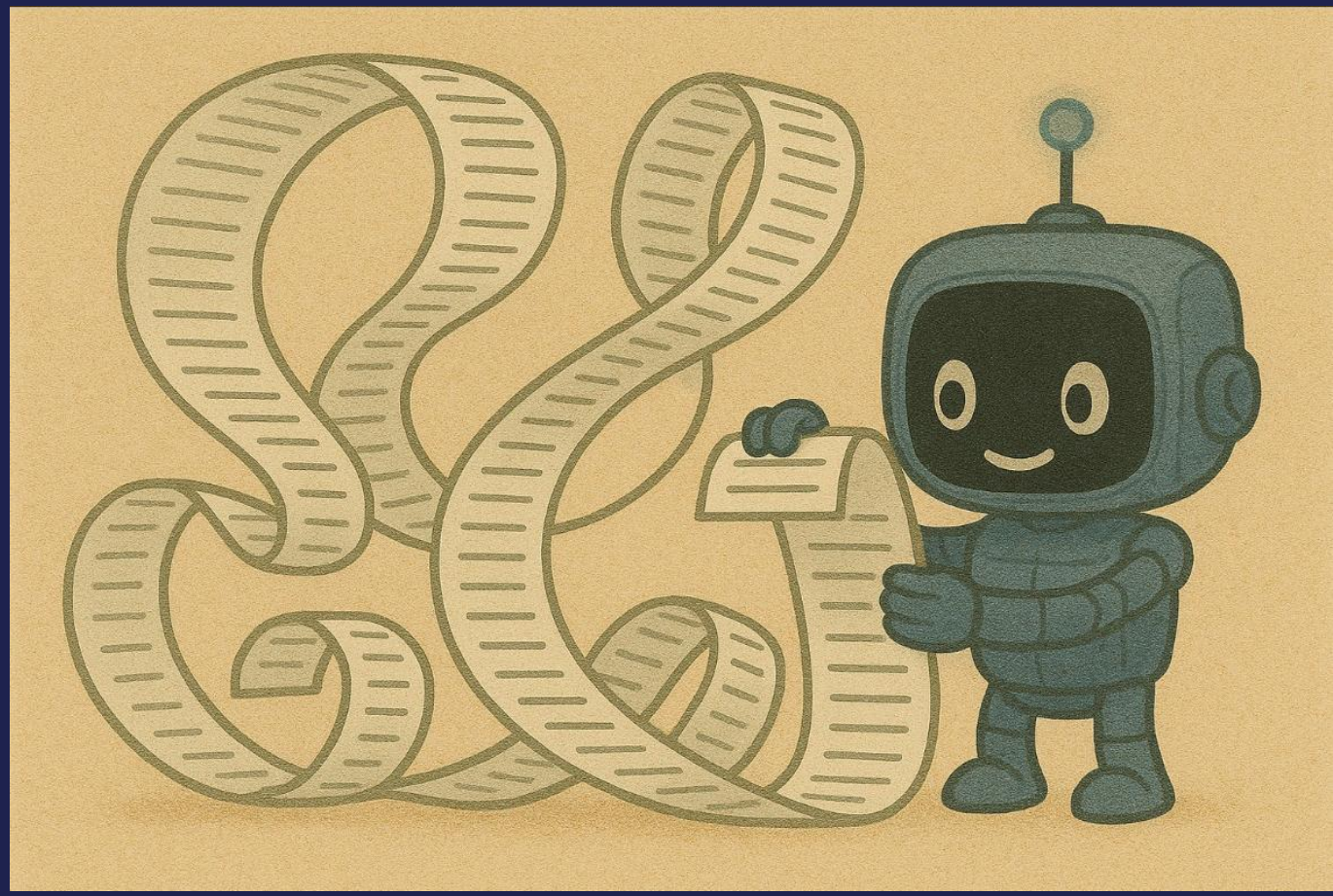
1. Create fusion teams (ML people, Devs, Business people) and come up with solutions to long-running projects or issues.

2. Build prototype fast and get into the hands to the Adopt crew

3. Report on progress in the next demo day

Make a wish list of where AI might help

1. Operations
2. Marketing
3. Sales
4. Finance
5. HR
6. IT
7. Many More



Create Grounding AI Agent Principles

1. Transparency

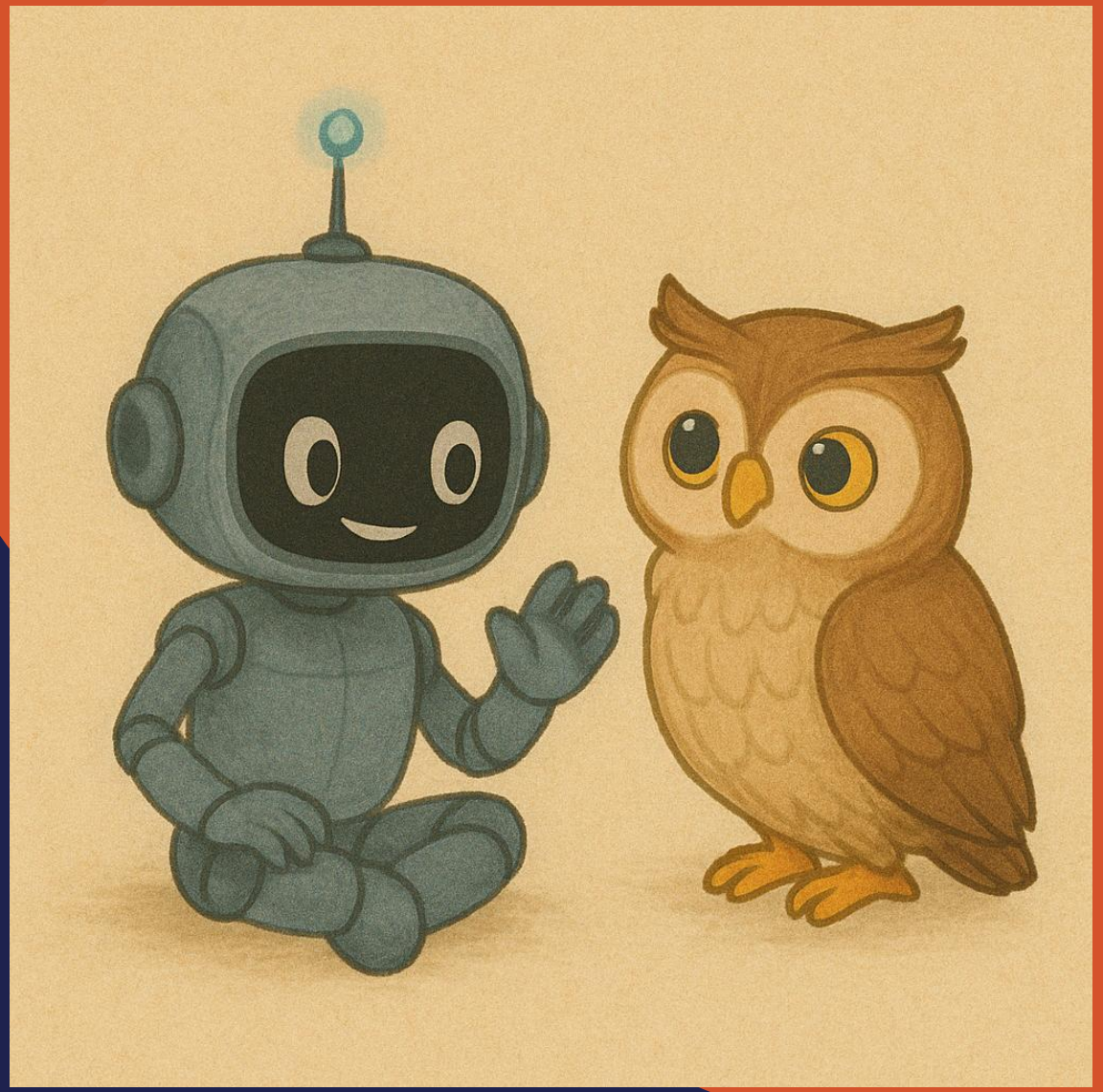
- Let the user know that the this is an AI-enabled Agent
- Provide some basic instructions on how to get started (e.g. a “Hello” message, sample prompts)
- Clearly document this on the product page
- Show the list of prompts a user has asked in the past.
- Make it clear how to give feedback (thumbs up and down, Send Feedback button, etc.).
- Clearly articulate if the Agent has usage or topic restrictions.

2. Control

- Make sure it’s clear how the user can modify the Agent after it’s been created with things like the System Prompt.
- Enable the user to choose how verbose the Agent is, its writing style, and any caveats on what the Agent should not talk about.
- Allow the user to view and delete any associated files or data, prompts, and past conversations.

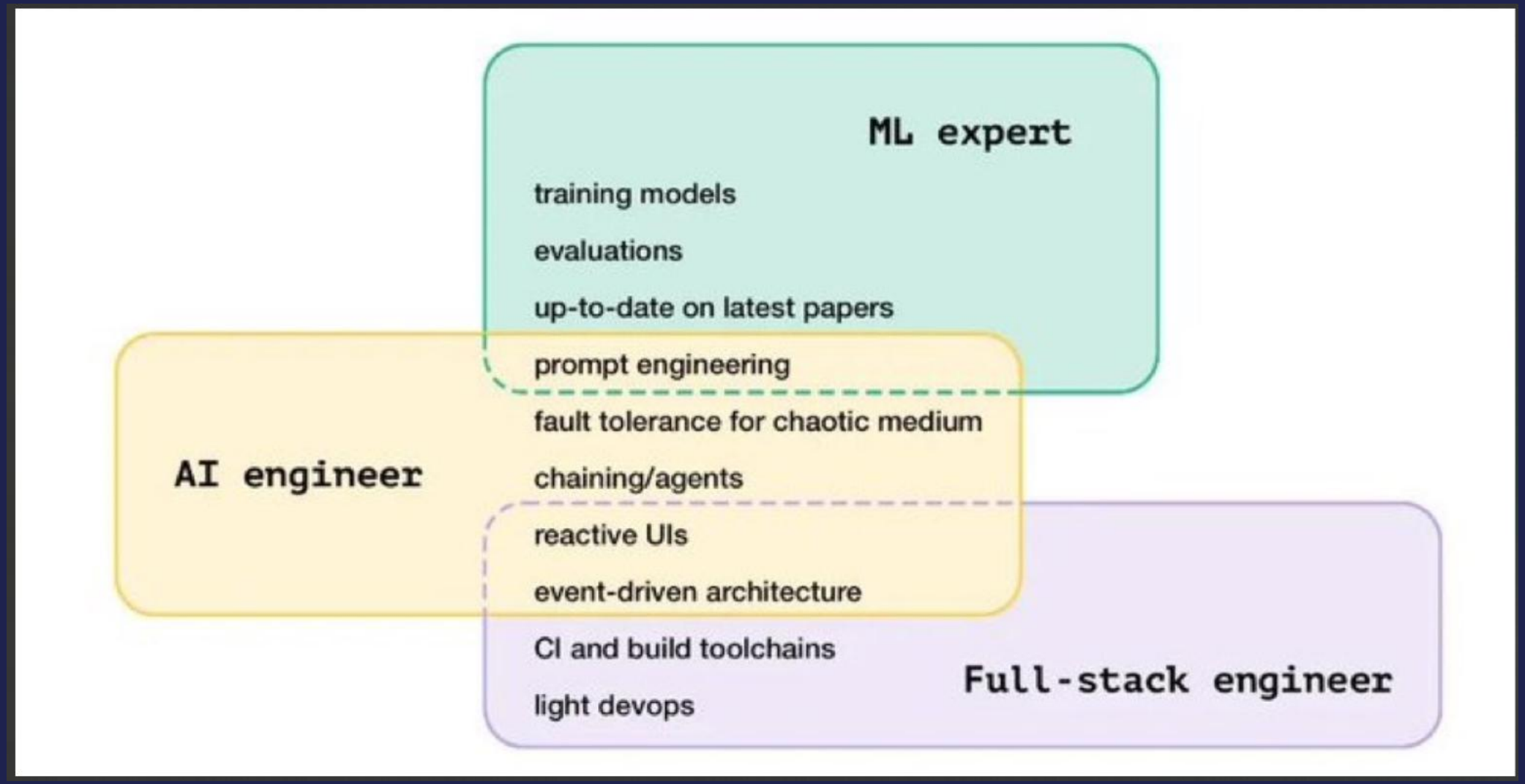
3. Consistency

- Make sure the icons for Share Prompt, add a file or photo and tag someone are standard and recognizable.
- Use the paperclip icon to indicate file upload/sharing with the Agent, and an image icon to indicate graphics upload.

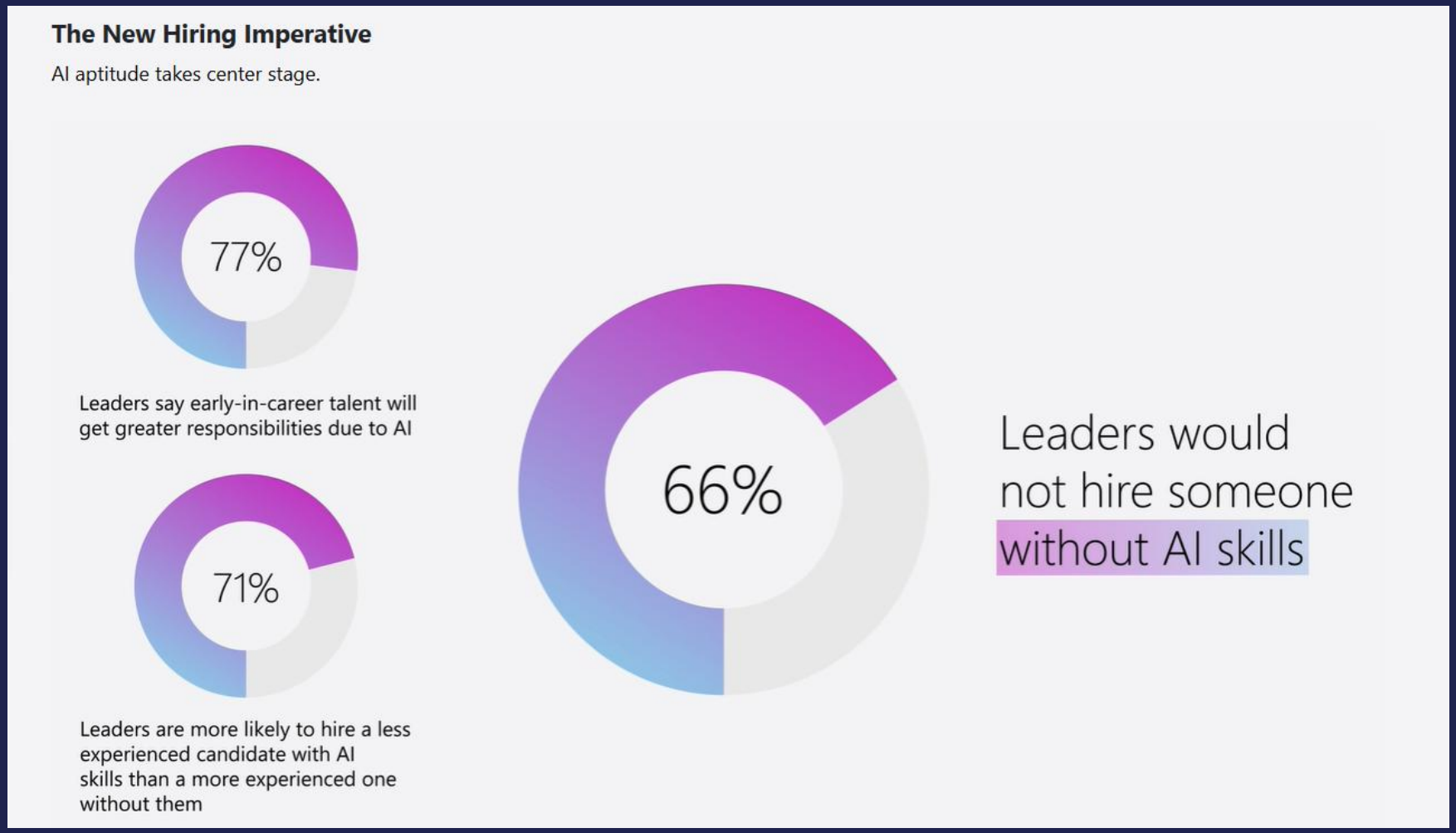


Part 5:

What does this mean for YOU?



AI Engineers are the most sought after people in tech right now



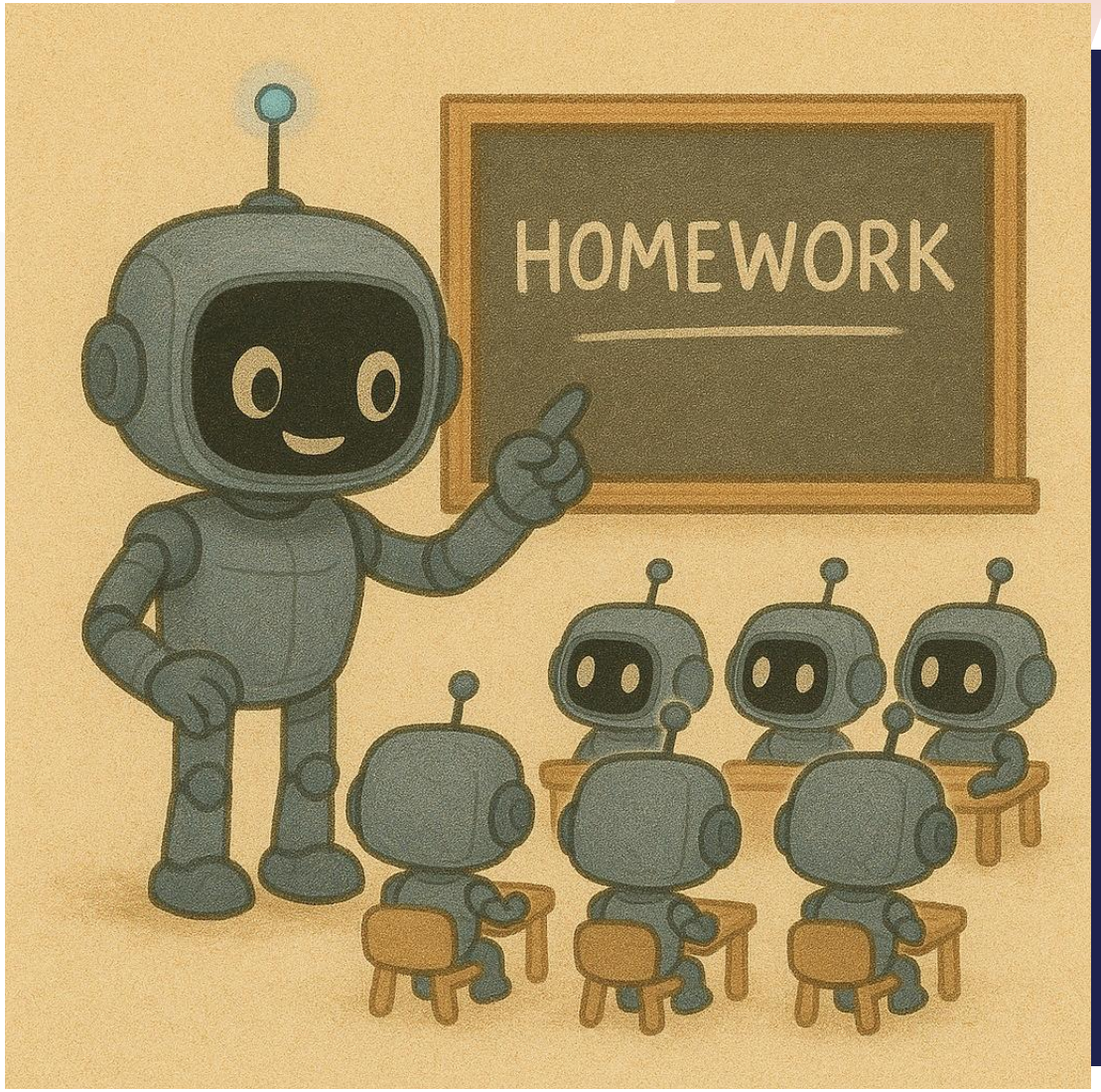
71% of leaders say that they'd hire

someone with AI skills > someone with experience

1. Use a wide variety of AI tools for your daily workflows

Document

Share



- Go to AI.Azure.com or copilotstudio.Microsoft.com
- Build something
- Join our AI community and share what you built!

Thank you

Try out the Azure AI Agent Service for free!

1. Join Discord
2. Join the channel #agentcon
3. Post a photo with #TryFoundryAgents
4. Follow the instructions in your DM



aka.ms/agentcon/discord